CSDWC: A Cross-Source Waveform Cooperation Method for Estimating Forest Canopy Height Change From GEDI and Airborne LiDAR-Simulated Waveforms

Huachen Zhao, Sheng Nie, *Member, IEEE*, Xiaohuan Xi, and Cheng Wang

***Abstract*—Integrating airborne light detection and ranging (LiDAR) with the global ecosystem dynamics investigation (GEDI) offers an effective way to characterize forest canopy height changes. However, the accuracy of conventional change estimation methods is often limited by systematic geolocation errors, intersensor inconsistencies, and signal coupling among canopy, ground, and noise components. To address these challenges, we develop a footprint-level canopy height change extraction framework by integrating GEDI observations and airborne LiDAR-simulated waveforms. First, a two-stage optimization algorithm that combines digital terrain model (DTM) matching and ground-return waveform matching is introduced to correct systematic geolocation errors in GEDI footprints. Second, a novel cross-source differential waveform cooperation (CSDWC) method is proposed to retrieve footprint-level canopy height changes by jointly analyzing GEDI waveforms and airborne LiDAR-simulated waveforms. This method reduces intersensor bias while improving the detection of weak canopy signals. Results show that the proposed two-stage optimization algorithm effectively corrects systematic geolocation errors, compensating for a mean absolute spatial offset of 10.07 m. The CSDWC method achieves reliable footprint-level canopy height change estimation over the five-year study period, with an overall root mean square error (RMSE) of 3.72 m and mean absolute error (MAE) of 2.07 m, representing reductions of 1.92 m (34.04%) in RMSE and 1.09 m (34.49%) in MAE compared with the conventional relative height (RH) metrics-based differencing approach. Overall, the proposed framework substantially improves the accuracy of forest canopy height change monitoring and provides a robust methodological basis for forest carbon stock and dynamics research.**

***Index Terms*—Airborne light detection and ranging (LiDAR), forest canopy height change, forest structure, geolocation correction, global ecosystem dynamics investigation (GEDI), multisource data integration, signal decoupling, waveform-based differencing, waveform simulation.**

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I. Introduction

FORESTS play a critical role in the global carbon cycle [1], climate regulation [2], [3], and ecosystem functioning [4] by controlling carbon sequestration, surface energy exchange, and biodiversity patterns. Forest canopy height is a fundamental structural attribute that is strongly associated with aboveground biomass [5], carbon stocks [1], and forest successional status [6]. Therefore, accurate monitoring of forest canopy height dynamics is essential for understanding ecosystem responses to climate change, natural disturbances, and human activities, which have increasingly altered forest structure across spatial and temporal scales [7], [8].

Field-based forest inventories provide reliable measurements of canopy height change, but their limited spatial coverage and temporal frequency make them unsuitable for large-scale and repeated monitoring. Remote sensing offers an effective alternative; however, different sensing modalities have inherent limitations. Optical imagery primarily captures horizontal canopy information and often suffers from signal saturation in dense forests, reducing its sensitivity to vertical structural variation [9]. Microwave remote sensing partially penetrates vegetation canopies but remains sensitive to terrain effects and signal saturation, particularly in high-biomass forests [10]. In contrast, light detection and ranging (LiDAR) directly measures vertical forest structure by recording the full vertical distribution of returned energy, making it particularly suitable for canopy height estimation and change detection [11], [12].

Airborne LiDAR systems (ALSs) can generate high-resolution and highly accurate canopy height models (CHMs) [13] and have been widely used for regional forest structure mapping [14], [15]. However, acquiring bi-temporal ALS data for large-scale change detection is frequently prohibitive due to exorbitant operational costs and intensive logistical requirements.

Space-borne LiDAR missions provide a complementary solution by enabling global-scale observations of forest vertical structure. The first-generation Ice, Cloud, and

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Land Elevation Satellite/Geoscience Laser Altimeter System (ICESat/GLAS) demonstrated the feasibility of space-borne waveform LiDAR but was constrained by sparse sampling and temporal discontinuity. Since 2018, second-generation missions, including ICESat-2/Advanced Topographic Laser Altimetry System (ATLAS) [16] and the global ecosystem dynamics investigation (GEDI), have substantially improved sampling density and data availability. GEDI, in particular, was specifically designed to characterize forest vertical structure using a multibeam full-waveform LiDAR system, covering latitudes from 51.6°N to 51.6°S [17]. However, despite these unprecedented capabilities, the relatively short operational period and geolocation uncertainty severely limit the availability of coincident overlapping footprints suitable for direct single-sensor change estimation [18]. Consequently, combining historical ALS data with contemporary GEDI observations provides a highly cost-effective alternative. This cross-sensor strategy enables large-scale bi-temporal monitoring utilizing only a single historical airborne acquisition and freely available space-borne data, thereby substantially reducing the financial and logistical burdens associated with repeated airborne campaigns.

Despite its immense practical value, extracting high-fidelity canopy height changes from ALS and GEDI data remains largely unexplored and methodologically challenging. Currently, the dominant paradigm relies on map-level differencing, in which independent wall-to-wall height maps are generated by fusing GEDI samples with multisource satellite imagery [19], [20], [21], [22], [23], [24], [25], [26] and then subtracted to estimate canopy height change [27], [28]. However, this map-to-map differencing strategy suffers from severe error propagation. Given that 92.1% of the total error in GEDI-based continuous canopy height mapping has been attributed to model upscaling [29], the accumulation of uncertainties and scaling biases across two epochs inevitably leads to pronounced degradation in change detection accuracy.

To avoid this compounding error, a more reliable paradigm shifts the focus from map-level differencing to direct change extraction at the footprint level. By accurately estimating canopy height changes at discrete GEDI footprints, this approach can generate high-fidelity training samples for subsequent continuous mapping, requiring only a single upscaling step and thereby avoiding the error accumulation associated with two-epoch predictions. However, accurate cross-sensor detection of canopy height change at the footprint level remains a critical gap in current research. This challenge is primarily constrained by three fundamental bottlenecks. First, GEDI observations are affected by systematic geolocation biases. Reported footprint offsets ranging from several to tens of meters [30], [31] may cause GEDI observations to sample heterogeneous forest conditions adjacent to their nominal locations, thereby degrading canopy height accuracy and propagating errors into change estimates. Second, the standard level 2A (L2A) relative height (RH) metrics exhibit intrinsic algorithmic limitations driven by severe signal coupling [32], [33]. Specifically, the coupling of weak canopy signals with background noise often causes the algorithm to miss the true canopy top [34], while the overlap of ground signals with lower-canopy returns introduces significant biases in ground elevation extraction [35]. This complex canopy-ground-noise signal coupling inevitably results in systematic deviations in the final estimation of canopy height. Third, substantial physical differences between ALS and GEDI, including footprint size, waveform characteristics, acquisition geometry, and noise properties, introduce systematic intersensor inconsistencies, making direct differencing of ALS-derived heights and GEDI L2A metrics unreliable for canopy height change detection.

To address these bottlenecks, this study proposes a novel cross-sensor framework for extracting high-fidelity, footprint-level forest canopy height changes. The foundation of this framework relies on large-footprint waveform simulation [36], which converts discrete-return ALS point clouds into GEDI-like waveforms to establish a harmonized physical baseline. Our primary objective is to retrieve accurate footprint-level canopy height changes by fusing a single historical airborne acquisition with contemporary GEDI observations. This single-epoch strategy eliminates the reliance on expensive bi-temporal ALS campaigns, providing a highly cost-effective paradigm. Furthermore, by jointly analyzing the simulated and observed waveforms, the proposed framework explicitly mitigates errors caused by GEDI's geolocation biases and the complex canopy-ground-noise signal coupling. The specific contributions of this study are twofold as follows.

1) A two-stage optimization strategy is proposed to correct systematic geolocation errors in GEDI footprints by combining digital terrain model (DTM) matching with waveform-based ground-return matching.
2) A novel cross-source differential waveform cooperation (CSDWC) method is introduced to extract canopy height change directly at the footprint level by differencing GEDI waveforms and airborne LiDAR-simulated waveforms, thereby mitigating intersensor bias and enhancing the detection of weak canopy signals that are commonly missed by the GEDI L2A algorithm.

The remainder of this study is organized as follows. In Section II, the study area and datasets used in this work are introduced. Section III describes in detail the principles and processes of the proposed method for estimating forest canopy height changes. The results are evaluated and analyzed in Sections IV and V. Section VI provides the conclusion and outlook of this study.

II. Materials

A. Study Area

The study was conducted at four NEON sites in the Southeastern United States: Dead Lake (DELA), Lenoir Landing (LENO), Jones Ecological Research Center (JERC), and Ordway-Swisher Biological Station (OSBS) (see Fig. 1). DELA and LENO are dominated by deciduous broadleaf forests and woody wetlands, featuring species such as sugarberry, American elm, and deciduous conifers like bald cypress. In contrast, JERC and OSBS are characterized by evergreen coniferous forests, primarily longleaf pine (*Pinus palustris*) savannas, interspersed with deciduous broadleaf communities such as turkey oak (*Quercus laevis*). Across these sites, the

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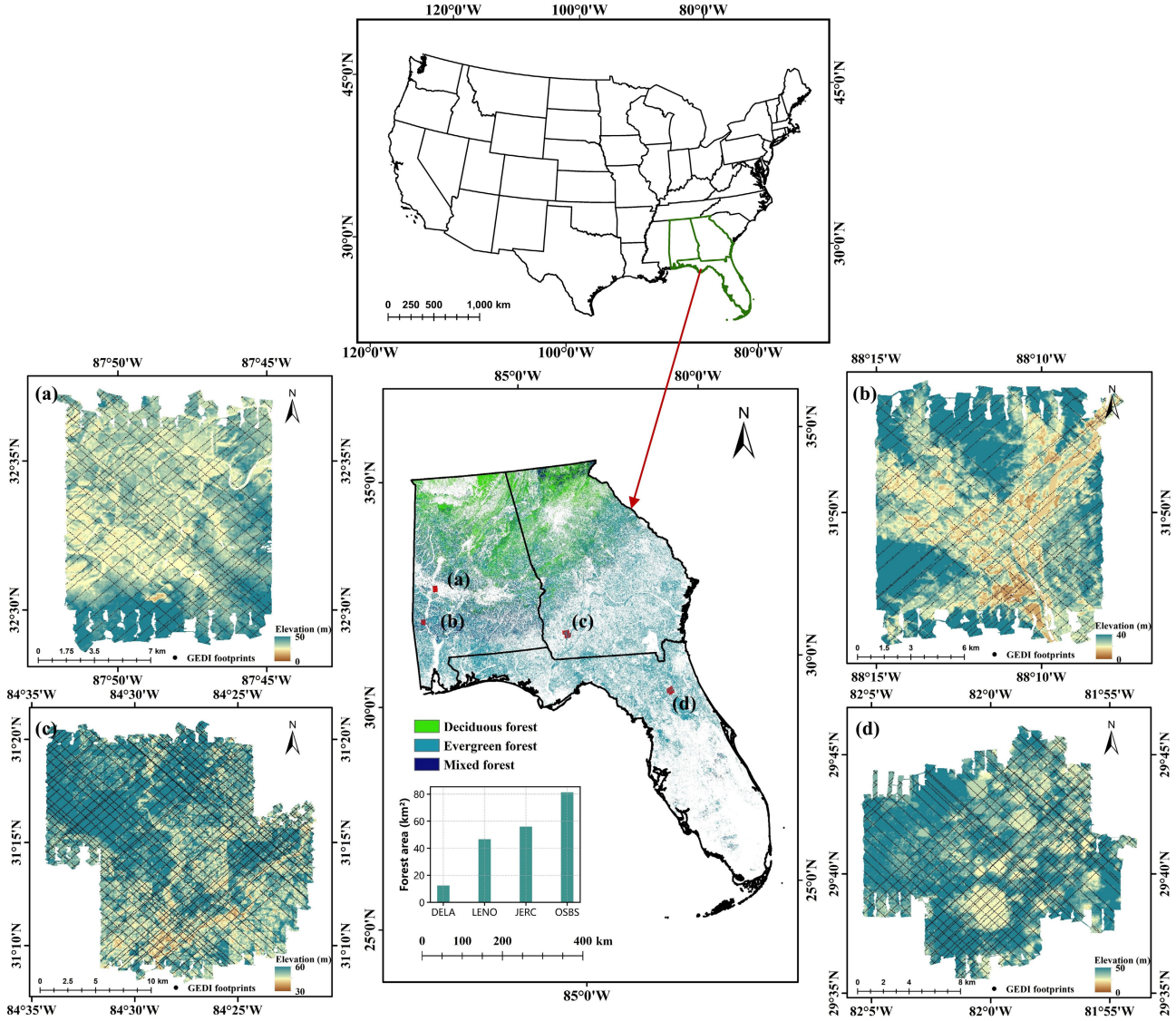


Fig. 1. Study area. (a)–(d) Correspond to the DELA, LENO, JERC, and OSBS sites, illustrating the spatial distribution of GEDI footprints.

mean annual temperature ranges from 17.6 °C to 20.9 °C, with annual precipitation between 1032 and 1385 mm; detailed site characteristics are summarized in Table I.

Collectively, these sites encompass a broad spectrum of forest ecosystems, ranging from deciduous broadleaf wetlands to evergreen coniferous savannas, in complex canopy structures and various environmental disturbances. All sites are covered by repeated NEON airborne remote sensing surveys, providing high-quality airborne LiDAR reference data. This combination of structural heterogeneity, varying disturbance histories, and consistent airborne LiDAR coverage makes these sites highly suitable for exploring the potential of footprint-level canopy height change extraction by integrating GEDI observations with airborne LiDAR-simulated waveforms.

B. GEDI Data

The GEDI mission is designed to characterize forest vertical structure and its temporal dynamics in support of carbon cycle and biodiversity research. The GEDI instrument employs three identical lasers operating at a wavelength of 1064 nm, emitting short pulses with a duration of 14 ns, a repetition frequency of 242 Hz, and a single-pulse energy of approximately 10 mJ [17]. Two lasers operate at full power, while the third coverage laser is split into two beams. These four beams are optically dithered to produce eight parallel ground tracks, with a nominal footprint diameter of approximately 25 m. GEDI observations cover latitudes from 51.6°N to 51.6°S.

In this study, GEDI level 1B (L1B) and L2A products acquired in 2021 were used. The L1B product provides geolocated full waveforms, while the L2A product includes processed waveform parameters, such as ground elevation and RH metrics that describe vertical forest structure. For each study site, the corresponding L1B and L2A datasets were obtained, including waveform profiles, footprint geolocation, elevation information, and RH metrics. All GEDI data

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TABLE I
DETAILED INFORMATION FOR EACH NEON SITE

Site	Area (km^2)	Mean elevation (m)	Slope range (°)
DELA	148.43	31.76	0 – 13.62
LENO	138.09	22.90	0 – 20.14
JERC	341.05	47.17	0 – 10.06
OSBS	222.12	38.43	0 – 8.45

TABLE II
TECHNICAL DETAILS OF THE AIRBORNE LASER SCANNING SENSOR AND FLYING PARAMETERS FOR THE STUDY SITES

Parameters	Values
Laser wavelength	1064 nm
Beam divergence	0.8 mrad
Flying altitude	1000 m AGL
Beam footprint size	∼0.8 m
Pulse repetition frequency	100 kHz
Point density	∼4 pts/m^2
Scan frequency	50 Hz
Data acquisition time	May 2016 and May 2021 (DELA, LENO)
	Sep 2016 and Sep 2021 (JERC, OSBS)

were downloaded from the NASA Earthdata platform (https://search.earthdata.nasa.gov).

C. NEON Airborne LiDAR Data

The NEON (https://data.neonscience.org) is a continental-scale, long-term monitoring project designed to unravel and forecast ecosystem responses to global climate shifts, land-use modifications, and biological invasions. NEON systematically conducts standardized remote sensing campaigns using ground-based sensors and high-resolution airborne observation platforms (AOPs) across diverse ecological domains. Key technical specifications of the ALS and flight parameters for each site are summarized in Table II.

In this study, we utilized the airborne LiDAR point clouds alongside their officially derived surface products, specifically the 1-m spatial resolution DTM and CHM [37]. The airborne LiDAR point cloud data acquired in 2016, along with the corresponding DTM, were used for algorithm development and waveform simulation, whereas the 2016 and 2021 CHMs were dedicated to deriving reference height changes for validation.

The horizontal coordinates of the NEON airborne LiDAR point clouds and their derived DTM and CHM products are referenced to the ITRF00 datum and projected into the Universal Transverse Mercator mapping frame. The elevations of the NEON ALS point clouds and DTM products are reported as orthometric heights referenced to NAVD88 using the GEOID12A geoid model. Accordingly, the elevations associated with the ALS-derived simulated waveforms are also referenced to NAVD88. In contrast, the CHM products are ground-normalized, with pixel values representing canopy heights above ground. To ensure a consistent vertical reference for elevation comparison and canopy height change estimation, the vertical coordinates of the GEDI waveforms were converted to NAVD88 before comparison with the ALS-derived simulated waveforms and DTM elevations.

D. NLCD Land Cover Data

Land cover information at a 30-m spatial resolution was derived from the National Land Cover Database (NLCD) (https://www.usgs.gov/centers/eros/science/annual-nlcd-data-access) for the years 2016 and 2021 to delineate forested areas and mask nonforest regions in GEDI observations. Specifically, the NLCD classification scheme characterizes forest cover into three distinct thematic classes: deciduous forest, evergreen forest, and mixed forest. These multitemporal land cover products provide a fundamental basis for delineating forested environments across the study area.

III. METHODS

A. Overview of the Methodological Framework

This study develops a fine-scale forest canopy height change extraction framework by integrating space-borne GEDI observations and airborne LiDAR-simulated waveforms. The overall workflow, illustrated in Fig. 2, comprises five primary components: 1) data filtering; 2) two-stage geolocation error correction; 3) intersensor bias correction via waveform simulation; 4) CSDWC height change extraction; and 5) accuracy assessment.

At the core of the framework is a waveform-based method that directly estimates canopy height change at the LiDAR footprint level. To ensure reliable waveform correspondence between GEDI and airborne LiDAR data, a two-stage optimization algorithm is first employed to correct systematic geolocation errors in GEDI footprints. Subsequently, a novel CSDWC method is introduced to retrieve footprint-level canopy height change by jointly analyzing GEDI waveforms and airborne LiDAR-simulated waveforms.

B. Data Filtering

GEDI footprints were filtered to ensure the reliability of subsequent analyses [18], [20], [38], [39]. First, footprints associated with invalid waveforms were removed ($quality_flag = 0$) [20]. Second, footprints were filtered using $degrade_flag = 0$ to exclude observations with potentially degraded geolocation or instrument conditions [18], [39]. Third, footprints with sensitivity ≥ 0.9 were retained to ensure sufficient laser energy penetration through forest canopies [20]. Fourth, footprints exhibiting elevation discrepancies greater than 50 m between GEDI-derived ground elevation and the reference NEON DTM were excluded, as such discrepancies are typically associated with cloud contamination or waveform anomalies [38].

In addition, land cover information was used to restrict the analysis to forested environments. The 30-m NLCD datasets were spatially intersected with GEDI footprints; a footprint was retained if its center fell within a pixel classified as

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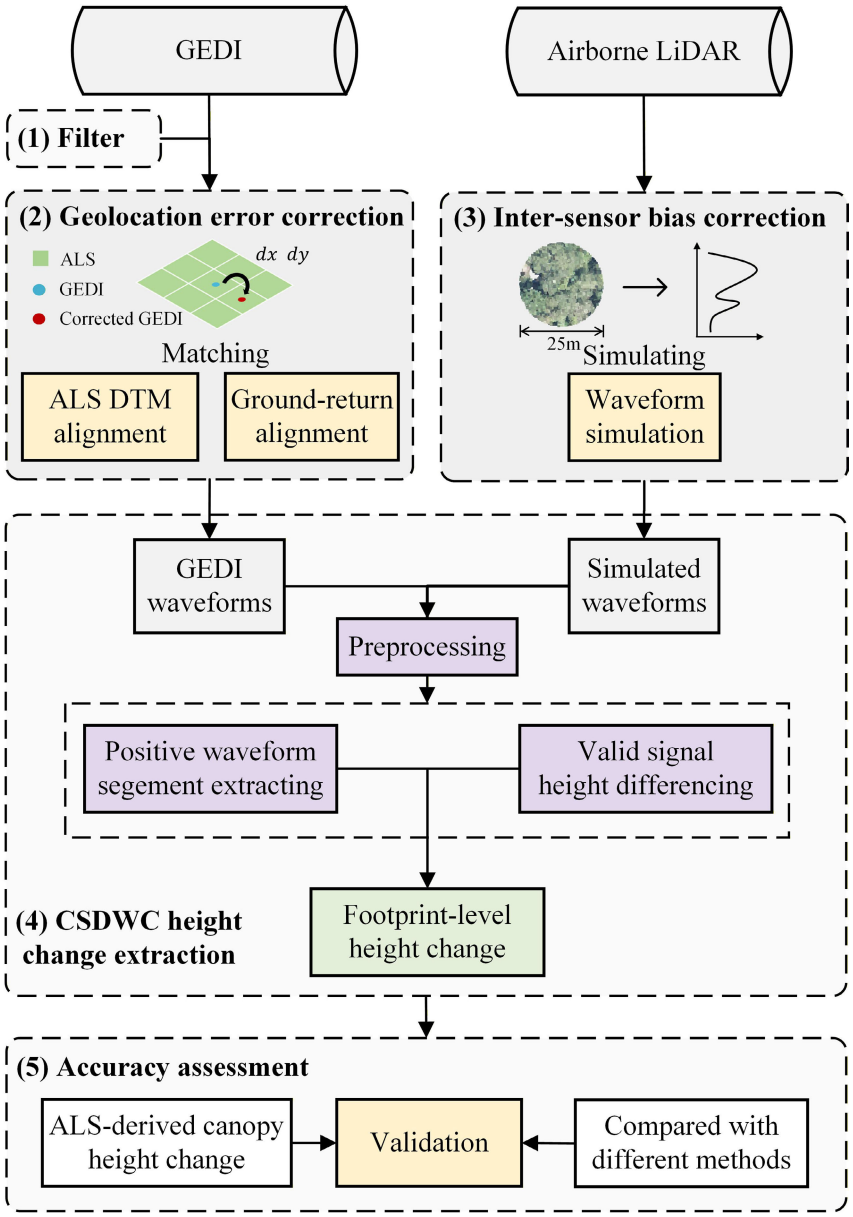


Fig. 2. Technical workflow for retrieving forest canopy height change at footprint-level.

deciduous, evergreen, or mixed forest in either 2016 or 2021. This inclusive masking strategy ensures that the analysis captures potential land-cover transitions, such as deforestation or reforestation, which occurred between the two observation epochs.

C. Two-Stage Geolocation Error Correction

Two primary strategies are commonly used for geolocation correction: waveform matching-based methods and DTM matching-based methods. Because the airborne LiDAR data used in this study were acquired approximately five years prior to the GEDI observations, forest structural changes may have occurred between the acquisition epochs. Consequently, ALS-derived simulated waveforms may deviate substantially from actual GEDI waveforms, rendering pure waveform-matching approaches unreliable. In contrast, DTM matching-based methods shift the underlying assumption from canopy consistency to terrain stability, formulating geolocation correction as a geometric optimization problem that aligns GEDI elevations with a reference DTM; however, these methods neglect the physical interaction between the GEDI waveform and the ground surface, thereby limiting correction accuracy.

To overcome these limitations, we propose a two-stage optimization strategy that integrates DTM information with airborne LiDAR-simulated ground-returns (see Fig. 3). This strategy offers two key advantages: 1) it eliminates the requirement for airborne LiDAR acquisitions co-temporal with GEDI and 2) it achieves higher correction accuracy than purely geometric approaches by incorporating physically consistent ground return information. The method consists

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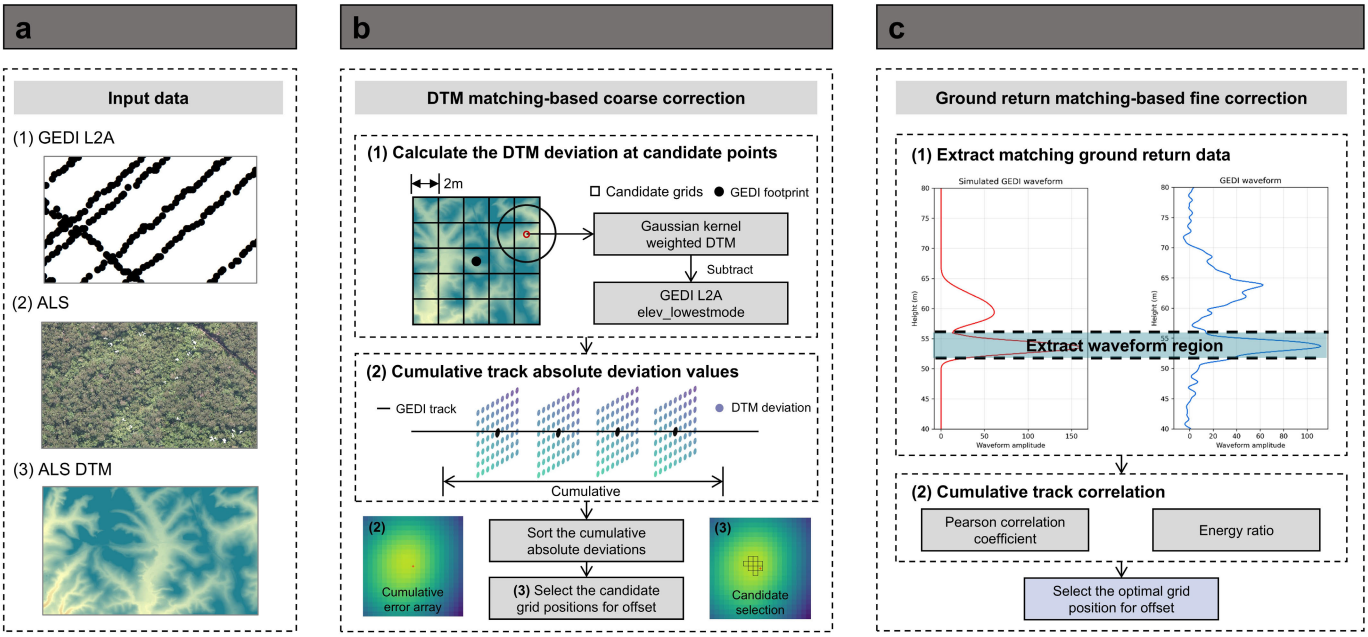


Fig. 3. Two-stage optimization method for geolocation bias correction: (a) input data; (b) DTM matching-based coarse calibration; and (c) ground-return matching-based fine calibration.

of two components: DTM matching-based coarse correction and ground-return matching-based fine correction. The coarse correction identifies candidate offset locations, while the fine correction determines the optimal offset from this candidate set.

1) DTM Matching-Based Coarse Correction: A grid-shifting approach was employed to generate candidate offset locations. Candidate footprints were produced by shifting the GEDI footprint center at 2-m intervals within a 40 × 40 m search window, resulting in 441 candidate positions. Following Tang et al. [40], geolocation offsets were assumed to be systematic at the track level. Although footprint-level correction is theoretically possible, it is often unstable in practice due to limited terrain variation and reprojection noise [40].

Following previous DTM matching-based GEDI geolocation correction studies that estimated footprint offsets by minimizing discrepancies between GEDI lowest-mode elevations and reference DTM elevations [40], [41], [42], for each candidate footprint, ALS-derived DTM values within a 12.5-m radius were extracted and convolved with a 2-D Gaussian kernel (σ = 5.5 m [40]) to approximate the spatial energy distribution of the GEDI footprint. The resulting weighted elevation was treated as the reference ground elevation. GEDI ground elevation was defined as the elevation of the lowest mode center (*elev_lowestmode*) and converted to the NAVD88 vertical datum. The absolute difference between the GEDI ground elevation and the simulated ground elevation was computed for each candidate footprint. A threshold of 15 m was applied to remove low-quality GEDI elevation estimates [40].

By aggregating elevation differences across all valid forest footprints within a track, a 3-D error array was constructed. Tracks containing fewer than 50 valid footprints were excluded to ensure robust estimation of the track-level offset, with this minimum sample-size threshold determined through the sensitivity analysis described in Section III-G. The grid cells were ranked by mean absolute elevation difference, and the top 10 candidates were retained for fine correction.

2) Ground-Return Matching-Based Fine Correction: Fine correction was performed by maximizing the similarity between ground-return segments of simulated airborne waveforms and actual GEDI waveforms for each candidate offset. For each candidate grid cell, ALS point clouds within a 12.5-m radius were extracted, and corresponding simulated waveforms were generated using the GEDI waveform simulator [36].

A wavelet-transform-based approach was used to delineate the ground-return segments because its localized multiscale representation is well suited to identifying coherent return pulses and their boundaries. By correlating the waveform with wavelet functions at different scales, return pulses with a continuous shape and finite width produce strong and spatially consistent responses, whereas residual small-amplitude fluctuations produce weaker or less persistent responses. In the nearly noise-free ALS-derived simulated waveforms, the ground return generally appears as a distinct low-elevation pulse, and its wavelet response can therefore be clearly distinguished from residual fluctuations. This enables the start and end positions of the ground-return segment to be identified reliably from the wavelet-enhanced response. Using the indices of these segments, the corresponding portions were extracted from the NAVD88-aligned GEDI waveforms. Similarity between simulated and GEDI ground-return segments was quantified using a combined metric consisting of the Pearson correlation coefficient and the energy ratio, yielding an integrated similarity score S_{int}

$$S_{\text{int}} = \frac{\sum_{i=1}^{N}(s_i-\bar{s})(g_i-\bar{g})}{\sqrt{\sum_{i=1}^{N}(s_i-\bar{s})^2}\sqrt{\sum_{i=1}^{N}(g_i-\bar{g})^2}} + \frac{\sum_{i=1}^{N} g_i^2}{\sum_{i=1}^{N} s_i^2} \quad (1)$$

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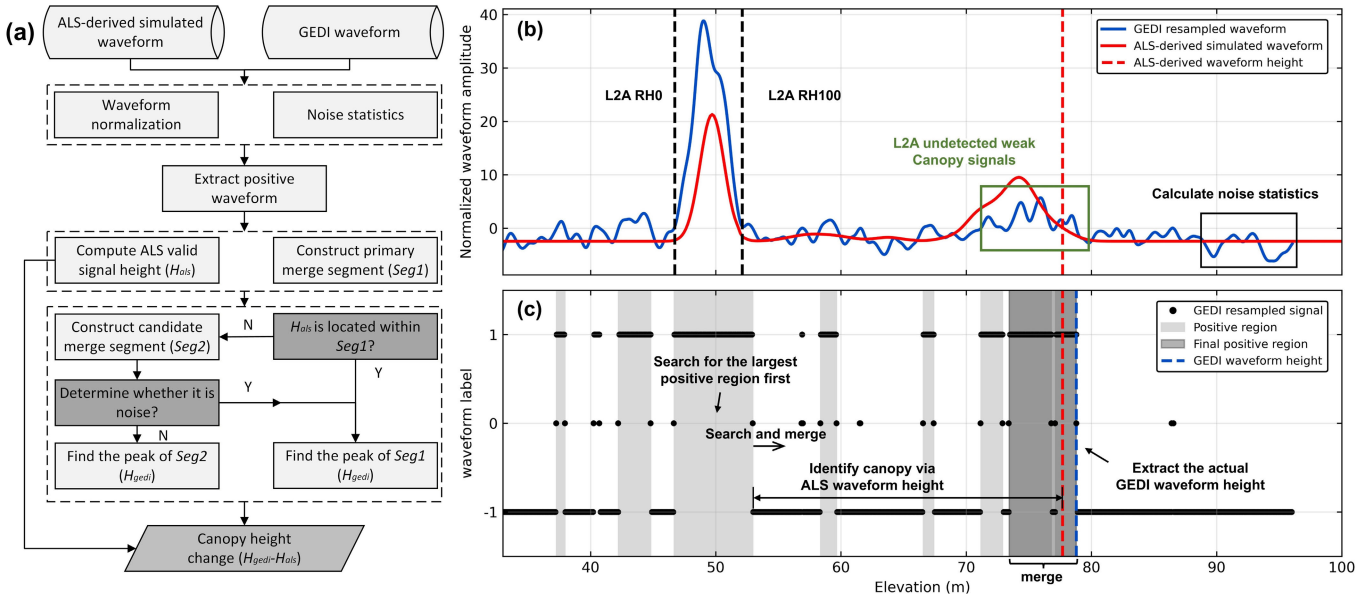


Fig. 4. CSDWC algorithm for canopy height change extraction: (a) overall workflow and (b) and (c) examples where GEDI substantially underestimates canopy height and the corresponding CSDWC correction.

where s_i and g_i denote the ith sample of the simulated and GEDI ground-return segments; $\bar{s}$ and $\bar{g}$ are their corresponding means; and N is the number of samples.

The first term captures waveform-shape similarity, whereas the second term reflects amplitude agreement, and their sum yields the overall similarity measure. The integrated similarity was computed for each candidate and averaged across all footprints in the track, producing a 3-D similarity surface. The grid cell with the maximum mean similarity was identified as the optimal offset location, and the associated displacement vector was applied as the systematic geolocation correction for the entire track.

D. Intersensor Bias Correction via Waveform Simulation

Systematic intersensor biases arise from differences in footprint size, acquisition geometry, and signal characteristics between GEDI and ALSs. To mitigate these effects, ALS point clouds were converted into GEDI-like waveforms through waveform simulation.

Using geolocation-corrected GEDI footprint centers as references, ALS point clouds within a 12.5-m radius were extracted. The GEDI simulator [36] was then applied with standard instrument configurations to generate simulated waveforms that reproduce GEDI footprint size, pulse shape, and vertical sampling characteristics. This procedure establishes a one-to-one correspondence between simulated and actual GEDI waveforms, enabling consistent waveform-level comparison.

E. CSDWC Height Change Extraction

To accurately retrieve canopy height change from GEDI and simulated waveforms, it is necessary to address the coupling among canopy, ground, and noise signals. In dense forests, strong canopy-ground coupling may lead to ground misidentification and systematic canopy height underestimation. In sparse forests or along forest edges, weak canopy returns are often confounded with background noise. Both situations degrade height retrieval accuracy and propagate errors into canopy height change estimates.

To address these issues, we propose the CSDWC algorithm (see Fig. 4) and compare its performance with the conventional RH metrics-based differencing approach, in which canopy height change is computed by differencing GEDI RH98 and ALS-derived simulated waveform RH98. Unlike traditional methods that estimate change by subtracting two independently retrieved canopy heights (each requiring canopy-top and ground detection), CSDWC directly derives canopy height change from the difference between the valid signal tops in GEDI waveforms and ALS-derived simulated waveforms. This method avoids explicit ground extraction by using the height-aligned simulated waveform as an external structural prior for ground referencing, thereby reducing errors induced by ground-canopy coupling. Furthermore, the joint use of ALS-derived simulated waveforms improves the discrimination of weak canopy signals embedded in noisy GEDI returns, especially under sparse-canopy or low-SNR conditions. The CSDWC workflow consists of three steps: 1) waveform preprocessing; 2) positive waveform segment extraction; and 3) valid signal height differencing.

1) Waveform Preprocessing: To suppress instrumental and atmospheric noise in GEDI waveforms, we characterized background noise using the mean and standard deviation of signal intensity within the top-most 100 bins of the waveform profile, denoted as Ω_{noise}, which represents the noise-dominated region. GEDI waveform intensity was adjusted by subtracting the mean noise level plus 1.5σ. To ensure waveform comparability, energy normalization was then applied by rescaling the simulated waveform intensity to match the GEDI waveform.

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The scaling factor was computed as the ratio of total signal energies ($totE$) between the two waveforms

$$totE = \Sigma_{i=1\ I_i>T} I_i \tag{2}$$

where I_i represents the intensity at the ith sample and the threshold T denotes the mean noise level.

Finally, cubic spline interpolation was applied to both GEDI and ALS-derived simulated waveforms to generate densely sampled profiles at 0.05-m vertical intervals.

2) Positive Waveform Segment Extraction: To isolate valid canopy and ground returns from noise-induced fluctuations, we identify continuous waveform intervals with positive intensity that persist for at least a minimum length L_{min}. The waveform samples are discretized into three labels: 1 (intensity > 0), 0 (intensity = 0), and −1 (intensity < 0). A consecutive sequence of label 1 is considered a potentially valid segment only if its length exceeds L_{min}.

The threshold L_{min} is derived from background noise statistics based on Rice's random process theory [43], serving as a robust criterion for rejecting transient noise spikes. Let $n(t)$ denote a noise sequence of length K extracted from Ω_{noise}. Since Ω_{noise} consists of the top-most 100 waveform bins, K is set to 100 in this study, with $t = 1, 2, \ldots, K$ representing the discrete index along the height coordinate. After detrending to ensure zero mean, the forward finite difference $\Delta n(t) = n(t+1) - n(t)$ is used to approximate the noise derivative. The zeroth- and second-order spectral moments are computed as

$$\sigma_0^2 = \frac{1}{K}\sum_{t=1}^{K} n(t)^2, \quad \sigma_1^2 = \frac{1}{K-1}\sum_{t=1}^{K-1} (\Delta n(t))^2 \tag{3}$$

where σ_0^2 characterizes noise amplitude variance (overall energy), and while σ_1^2 quantifies the variance of the noise derivative (high-frequency roughness).

Following Rice [43], the minimum valid segment length is defined as

$$L_{min} = \left\lceil \pi \sqrt{\frac{\sigma_0^2}{\sigma_1^2}} \right\rceil \tag{4}$$

where $\lceil \cdot \rceil$ is the ceiling operator. Conceptually, L_{min} represents an upper bound on the typical duration of contiguous noise excursions; segments longer than L_{min} are statistically unlikely to be produced by random noise alone and are therefore treated as potentially valid returns.

3) Valid Signal Height Differencing: Canopy height change is computed as the difference between the valid signal tops of the GEDI waveform and the ALS-derived simulated waveform. The valid signal top is defined as the highest elevation with intensity greater than zero under the idealized assumption of a noise-free waveform. Because simulated waveforms are effectively noise-free, their valid signal tops can be obtained directly. In contrast, GEDI waveforms are affected by residual noise even after suppression, making signal-top identification more challenging. To improve robustness, we adopt a bottom-up vertical search strategy that first delineates reliably detectable signal regions through segment merging, and then applies conditional rules using the simulated valid signal top as a soft constraint to recover canopy returns in noise-affected regions.

After discretization, positive signal segments are integrated using a connectivity-based merging strategy. First, the positive waveform segment with the largest vertical extent is selected as the initial segment, and neighboring segments are iteratively merged in the upward direction within a spatial range defined by the adaptively derived search distance, D_{search}, which characterizes the effective correlation length of the waveform in a statistical sense and is adaptively derived from the normalized autocorrelation function (ACF).

Let $S(t)$ denote the waveform amplitude at sample index t, which can be regarded as a superposition of multiple Gaussian echo components [44]. The normalized ACF $\rho(\tau)$ was computed as follows:

$$\rho(\tau) = \frac{\sum_t S(t)\, S(t+\tau)}{\sum_t S(t)^2} \tag{5}$$

where τ is the discrete lag (in units of samples) and is computed by shifting the waveform by integer sample steps. Specifically, for each $\tau = 0, 1, 2, \ldots,$ $S(t+\tau)$ is the waveform shifted forward by τ samples relative to $S(t)$, and the summation is performed over all overlapping points, thereby quantifying the similarity between the waveform and its shifted version at different lags.

At small lags, the ACF is primarily dominated by the autocorrelation of individual Gaussian echo components, while the cross correlation contributions between different echoes are negligible. Therefore, the decay behavior of its main lobe can be used to characterize the effective correlation length at the scale of a single echo. In this study, the smallest positive lag at which the ACF first decays to the $1/e^2$ is defined as the characteristic lag radius $R_{ACF(1/e^2)}$, as shown in the following equation:

$$R_{ACF(1/e^2)} = \min\left\{\tau > 0 \mid \rho(\tau) \le e^{-2}\right\}. \tag{6}$$

To account for the intrinsic broadening introduced by the autocorrelation operation, which increases the effective correlation width by a factor of $(2)^{1/2}$, and to convert the lag radius into an equivalent diameter, the search distance D_{search} is defined as follows:

$$D_{search} = 2 \times \frac{1}{\sqrt{2}} R_{ACF(1/e^2)} = \sqrt{2} R_{ACF(1/e^2)}. \tag{7}$$

If another positive segment is found within this search distance, it is merged with the current segment to form an expanded segment. This expanded segment becomes the new seed, and the merging proceeds iteratively until no additional segments fall within D_{search}. The resulting segment is termed the primary merged segment. This is motivated by the observation that, after noise suppression, the largest positive segment typically corresponds to ground return, canopy return, or a mixed ground-canopy signal, whereas smaller segments often represent residual noise or sparse canopy components.

The GEDI valid signal top is then determined using the simulated waveform top as prior information under the following rules. First, if the simulated top lies within D_{search} of the primary merged segment, the highest point of the primary merged segment is taken as the GEDI valid signal top. Otherwise, a missed detection due to weak canopy response

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is assumed. In this case, an upward search is initiated starting from a position $2 \times D_{search}$ below the simulated top. If a positive segment is detected within this search range, it is used as a new seed, and the same neighborhood search and iterative merging procedure is applied to obtain a candidate merged segment. To reduce noise contamination, the highest elevation of the candidate merged segment is compared against the highest elevation of any positive segment located to the left of the ground-echo peak in the airborne waveform (or 0 if none exists), excluding the segment containing the peak itself. If the candidate segment exhibits a higher amplitude, its top is adopted; otherwise, the primary merged segment top is retained. The resulting top then serves as the GEDI valid signal top. Finally, canopy height change is computed as the difference between the GEDI and simulated valid signal tops.

F. Accuracy Assessment

The accuracy evaluation comprised three components. First, we verified the feasibility of the CSDWC method under idealized conditions using bi-temporal simulated airborne waveform data. Specifically, we randomly selected 4595 footprints encompassing diverse forest structures at the LENO site and calculated the canopy height changes from these simulated waveforms using the CSDWC method. Because the footprint data were generated with identical sensor parameters and zero geolocation error, we could effectively eliminate the confounding effects of potential sensor biases and terrain-induced waveform broadening on height change extraction. This controlled setup allowed us to explore the core mechanism of the method: verifying whether the vertical shifts in the highest detectable waveform signal are primarily driven by true canopy structural changes.

Second, a footprint-level validation of CSDWC-derived canopy height changes was conducted against ALS references. To match the GEDI footprint, 1-m resolution ALS CHM pixels within a 12.5-m radius of each footprint center were extracted, and their 98th percentile was calculated. After the data filtering described above, a total of 6186 valid GEDI footprints with corresponding ALS references were retained across the four study sites, including 356, 742, 3332, and 1756 footprints at DELA, LENO, JERC, and OSBS, respectively. Method performance was evaluated using root mean square error (RMSE), R^2, bias, and mean absolute error (MAE). These metrics are computed as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i)^2} \tag{8}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(X_i - Y_i)^2}{\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2} \tag{9}$$

$$bias = \frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i) \tag{10}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|X_i - Y_i| \tag{11}$$

where X_i denotes the ith canopy height change estimated by the proposed method, Y_i represents the corresponding ALS-derived canopy height change, $\overline{Y}$ is the mean of the ALS-derived changes, and n is the number of samples.

Finally, the CSDWC method was compared with conventional RH metrics-based differencing strategies, in which canopy height changes were calculated by subtracting the ALS-derived simulated-waveform RH98 from the corresponding GEDI RH98.

G. Track Sample-Size Sensitivity for DTM Matching-Based Coarse Correction

To determine the minimum track sample size required for the DTM matching-based coarse correction, we conducted a trial-and-error sensitivity analysis using 13 candidate sample sizes: 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, and 200 valid footprints. Only GEDI tracks containing at least 200 valid footprints were included in this controlled sensitivity analysis. A total of eight GEDI tracks satisfied this criterion, allowing all candidate sample sizes to be evaluated using the same pool of eligible tracks.

For each sample size, 100 track-level trials were conducted. A sequence of 100 track selections was first generated from the eligible-track pool, with repeated selection of the same track permitted. The identical track-selection sequence, including the frequency and order of repeated tracks, was reused for all candidate sample sizes to control for track-to-track variability. In each trial, the specified number of valid footprints was randomly selected from the corresponding track and used to estimate the track-level offset through the DTM matching-based coarse correction. Correction performance was quantified by the reduction in canopy-height RMSE after applying the coarse correction relative to the RMSE before correction.

IV. Results

A. Results of Two-Stage Geolocation Error Correction

Fig. 5(a)–(d) illustrates the geolocation correction offsets for GEDI L2A tracks across the four study areas. A total of 13 tracks were processed using the two-stage optimization approach. During the DTM matching-based coarse correction, track-level offsets exhibited a clear unimodal distribution, with a peak between 10 and 15 m and a mean offset of 10.82 m. After applying the fine correction based on ground-return waveform matching, the mean track-level offset decreased to 10.07 m, and 54% of the footprints were concentrated within the 0–5-m offset range.

Table III summarizes the effects of geolocation correction on canopy height change estimation accuracy. For canopy height change estimation, both correction stages resulted in more pronounced improvements. On average, coarse correction reduced RMSE and MAE by 0.20 m (4.46%) and 0.22 m (8.61%), and fine correction further reduced them by 0.48 m (10.98%) and 0.33 m (12.80%). While the coarse correction consistently improved estimation accuracy for the DELA, JERC, and OSBS sites, it temporarily degraded the performance at the LENO site, increasing the RMSE from

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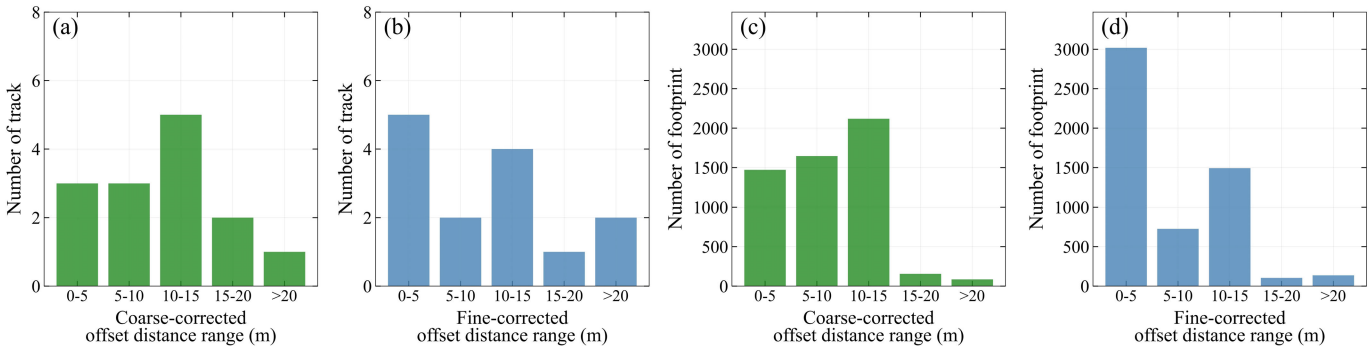


Fig. 5. Results of systematic geolocation error correction using the two-stage optimization algorithm. (a)–(d) Statistics of track-level and footprint-level offsets after coarse correction and fine correction.

TABLE III
COMPARISON OF CANOPY HEIGHT CHANGE ESTIMATION ACCURACY UNDER DIFFERENT PROCESSING WORKFLOWS, THE BEST RESULTS ARE MARKED IN BOLD BLACK

Site	Processing	Canopy height change		
		RMSE (m)	MAE (m)	Bias (m)
DELA	Unadjusted	6.26	3.28	0.13
	Coarse correction	5.68	2.75	**0.06**
	Fine correction	**5.31**	**2.59**	0.08
LENO	Unadjusted	3.79	2.61	-1.80
	Coarse correction	4.11	2.73	-1.91
	Fine correction	**3.68**	**2.61**	**-1.79**
JERC	Unadjusted	4.65	2.51	-0.16
	Coarse correction	4.29	2.22	-0.17
	Fine correction	**4.18**	**2.20**	**-0.15**
OSBS	Unadjusted	2.79	1.82	0.45
	Coarse correction	2.63	1.64	0.48
	Fine correction	**2.40**	**1.51**	**0.40**

3.79 to 4.11 m. However, the waveform-based fine correction successfully compensated for this intermediate degradation, reducing the final RMSE at LENO to 3.68 m (outperforming the unadjusted baseline). This confirms that footprint-level structural matching acts as an essential compensatory mechanism when track-level macro-terrain matching introduces local misalignments, ensuring robust height change retrieval. Overall, these results demonstrate the effectiveness of the proposed two-stage optimization algorithm and underscore the importance of geolocation correction in footprint-level canopy height change retrieval.

B. Impact of Intersensor Bias Correction

The impact of inherent discrepancies between GEDI and ALSs was quantified by examining the height differences among GEDI waveform RH98, ALS-derived simulated waveform RH98, and ALS-derived canopy heights [see Fig. 6(a)]. Quantitative results demonstrate that both the GEDI-derived and simulated waveform heights systematically underestimated the discrete-return airborne LiDAR measurements, exhibiting median deviations of 1.68 and 0.63 m. This shared tendency of underestimation reflects the physical differences between large-footprint full-waveform observations and high-density discrete-return scanning.

Consequently, if GEDI observations were directly compared to ALS-derived CHMs, these intersensor discrepancies would be erroneously interpreted as actual forest structural changes. By projecting the historical ALS data into the same observational domain as GEDI via waveform simulation, the systematic biases introduced by cross-platform inconsistencies were effectively neutralized. Specifically, as depicted in Fig. 6(b) and (c), the intersensor consistency correction reduced the RMSE by 0.35 m (5.84%) and attenuated the bias by 1.07 m (35.67%). Overall, this alignment establishes a consistent physical baseline, ensuring the accurate retrieval of actual canopy height changes.

C. Simulated Performance of CSDWC Method

The outcomes of the controlled bi-temporal waveform simulation at the LENO site (see Fig. 7), comprising 4595 randomly selected footprints encompassing diverse canopy structures and stand densities, yielding an R^2 of 0.94, an RMSE of 1.54 m, an MAE of 1.09 m, and a bias of −0.88 m. Under these ideal, noise-free, and spatially aligned conditions, this simulation establishes the theoretical upper bound for the CSDWC approach. It verifies that vertical shifts in the highest detectable waveform signal are primarily driven by true canopy structural changes when sensor and terrain artifacts are strictly controlled.

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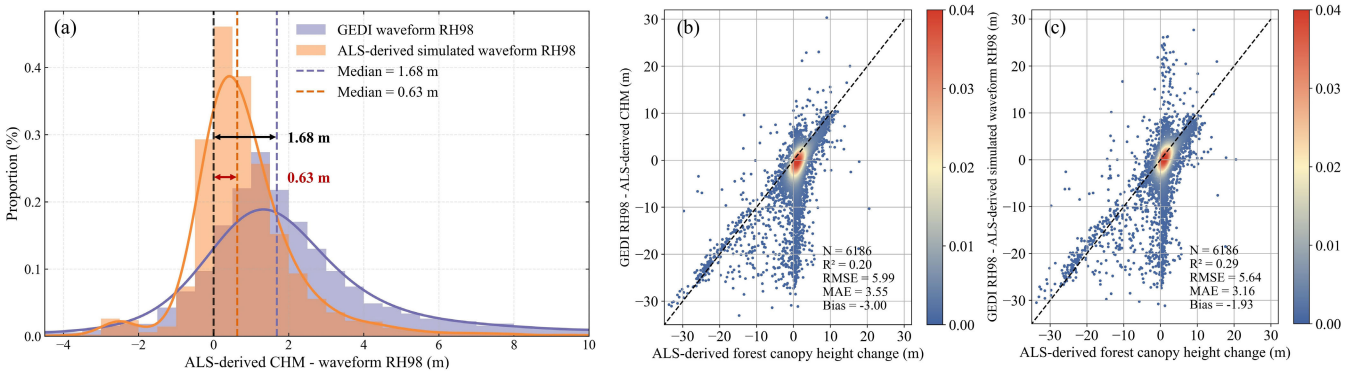


Fig. 6. Intersensor bias correction results: (a) RH98 deviations of airborne waveform simulations and GEDI relative to ALS-derived canopy height; scatter plots of height differences versus ALS canopy height changes demonstrating (b) uncorrected consistency using ALS-derived CHM and (c) corrected consistency using simulated GEDI-like waveforms.

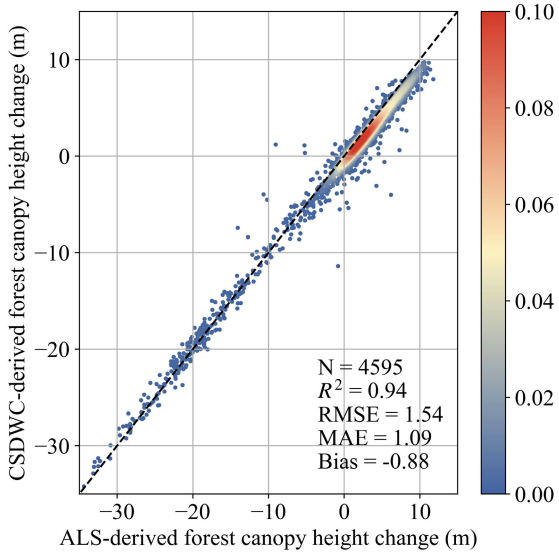


Fig. 7. Comparison between forest canopy height changes derived from the CSDWC method and ALS-derived reference change.

D. Validation of Footprint-Level Canopy Height Change Estimation

As illustrated in Fig. 8(a)–(d), the proposed CSDWC algorithm demonstrates consistent structural alignment with the ALS-derived reference data across different study sites. The footprint-level estimations generally aggregate along the 1:1 reference line, with the RMSE ranging from 2.40 to 5.31 m and the MAE spanning from 1.51 to 2.61 m. Furthermore, the overall bias is constrained within an interval from −1.79 to 0.40 m. As shown in Fig. 8(c), the algorithm maintains robust performance even in cases of substantial canopy height reduction (e.g., >10 m). Although some outliers persist, these discrepancies are likely attributable to residual GEDI geolocation uncertainties, which may cause the ALS-derived reference data to not perfectly align with the actual GEDI footprint coverage. These quantitative metrics indicate that the CSDWC method can effectively capture canopy height changes directly from waveform dynamics.

E. Comparison With RH Metrics-Based Differencing Strategies

Fig. 9(a) and (b) compares the accuracy of footprint-level canopy height change estimates obtained using the conventional RH metrics-based differencing approach and the proposed CSDWC method across the four study sites. Across the entire dataset, the conventional metrics-based approach exhibited a severe systematic shift, characterized by an overall RMSE of 5.64 m and a global average bias of −1.93 m. By transitioning to the CSDWC framework, this systematic underestimation was significantly mitigated. On average, the proposed method reduced the global RMSE by 1.92 m (34.04%) to 3.72 m, decreased the MAE by 1.09 m (34.49%) to 2.07 m, and narrowed the overall bias magnitude by 94.30% to a near-zero average of 0.11 m.

Furthermore, Fig. 9(c)–(e) provides representative waveform cases that illustrate the typical errors inherent in the standard RH metrics and the corresponding corrections applied via the CSDWC method. These profiles visually expose three primary waveform-level ambiguities responsible for severe height truncations: ground elevation estimation error, canopy incomplete detection, and canopy nondetection. As visualized in these case studies, the CSDWC algorithm effectively resolves these ambiguities by leveraging simulated baseline priors to firmly anchor the true signals. The underlying physical mechanisms driving these specific observation errors, along with the detailed rationale for the algorithmic corrections, are comprehensively explored in Section V-B.

F. Sensitivity Analysis

This sensitivity analysis examined how the number of valid GEDI footprints used to estimate the track-level offset affects the performance and stability of the DTM-matching-based coarse correction. As shown in Fig. 10, correction estimates based on small sample sizes, particularly 5–30 footprints, exhibit substantial variability and include cases with negative canopy-height RMSE reductions. This indicates that a limited number of footprints may be insufficient to provide a stable estimate of the track-level offset in the DTM-matching-based coarse correction.

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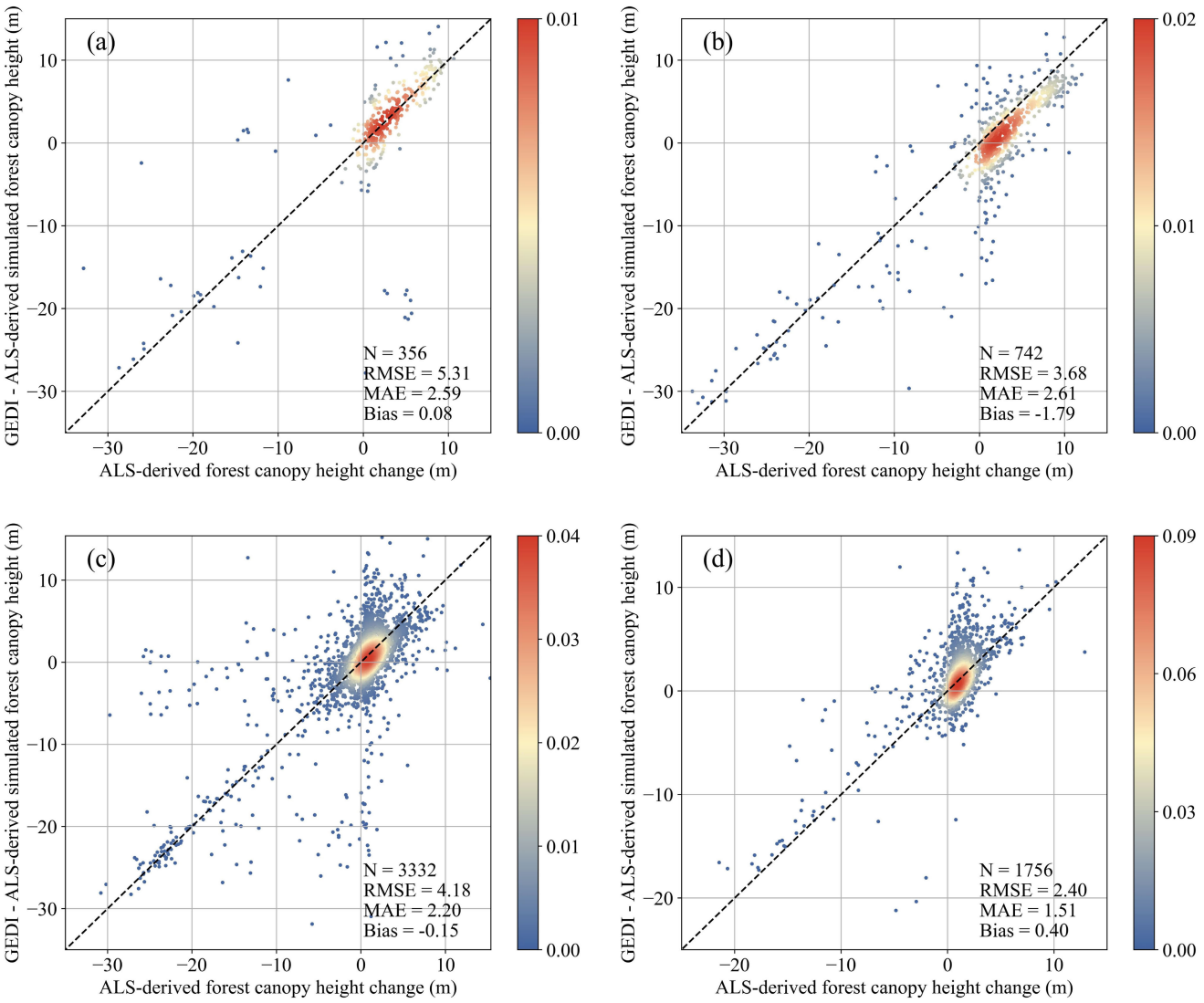


Fig. 8. Validation of footprint-level canopy height change extraction. (a)–(d) Scatter plots of CSDWC-derived against ALS-derived canopy height changes across the DELA, LENO, JERC, and OSBS sites.

When the number of selected footprints reaches 50, the variability of the correction results decreases noticeably, the median canopy-height RMSE reduction remains positive, and the resulting distribution is substantially narrower than the distributions obtained using smaller sample sizes. Increasing the sample size from 50 to 200 footprints further reduces the variability but produces only limited changes in the median RMSE improvement. Therefore, within this controlled sensitivity analysis, 50 valid footprints were selected as the practical minimum sample size for achieving stable DTM-matching-based coarse correction.

V. Discussion

A. Factors Modulating of CSDWC Method

The physical validity of retrieving canopy height change from large-footprint LiDAR depends on accurately decoupling actual structural dynamics from instrumental and topographic confounders. As characterized in previous studies, the transmitted pulsewidth inherently dictates the temporal broadening of the waveform [44], while terrain relief, such as slope and surface roughness, induces additional geometric spreading [35]. Both factors can precipitate apparent shifts in the canopy-top position that are entirely independent of actual vegetation growth or loss.

While our controlled simulations (see Section IV-C) theoretically validate the effectiveness of the CSDWC method, real space-borne GEDI observations inevitably involve additional practical complexities. Notably, residual geolocation uncertainty [31] and instrumental noise [34] can further distort waveform shape and obscure precise identification of the signal top. However, these inherent observational limitations do not undermine the conceptual validity of the waveform differencing paradigm. Rather, they highlight the necessity of the proposed analytical framework. By synergizing the two-stage geolocation optimization strategy with the CSDWC algorithm, our approach explicitly mitigates these geometric and radiometric distortions, ensuring that authentic canopy height changes can be reliably retrieved from real-world GEDI data even under challenging observation conditions.

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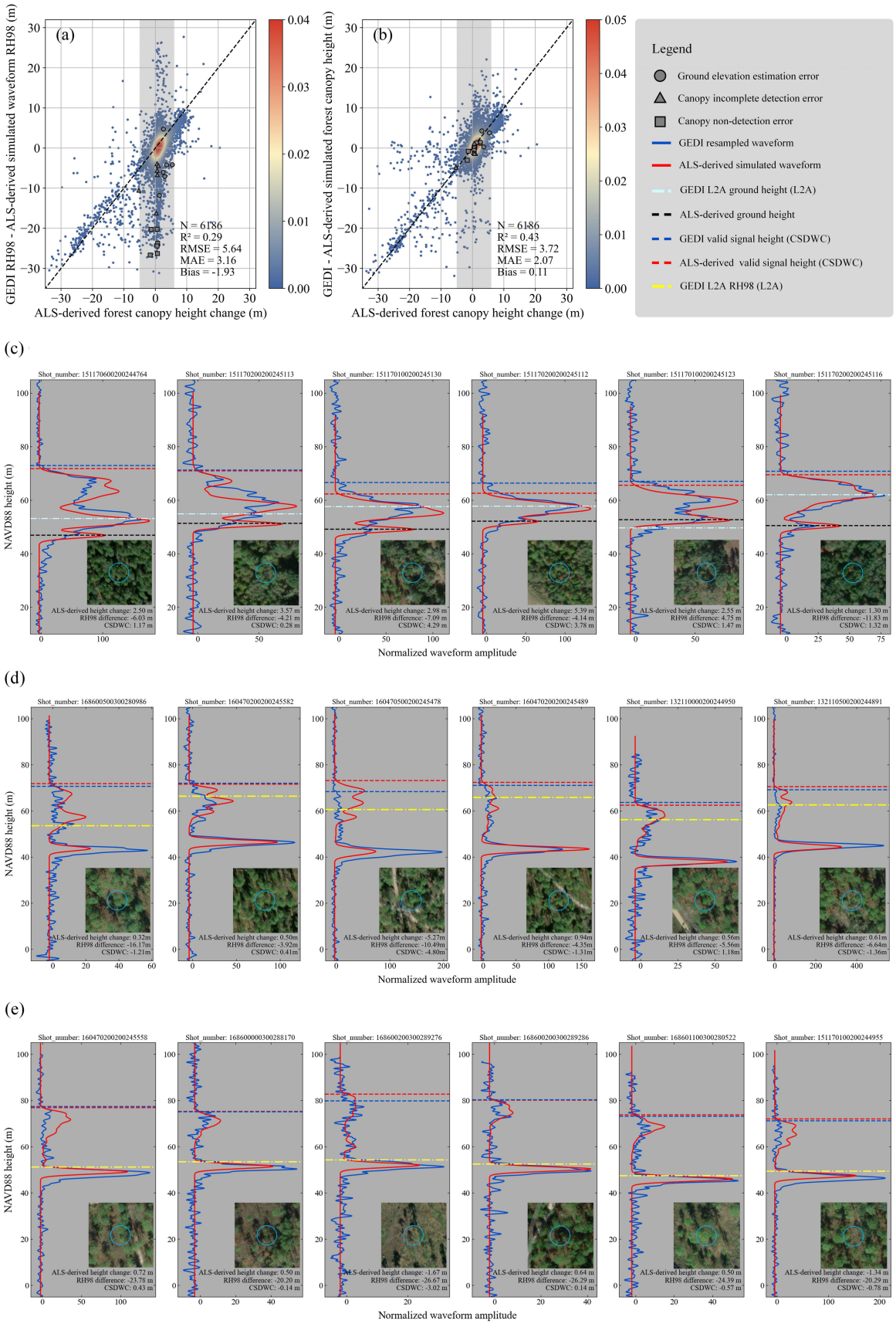


Fig. 9. Demonstration of performance improvements achieved by the CSDWC method in canopy height change extraction: (a) RH metrics-based differencing results; (b) CSDWC-based results; and (c)–(e) representative waveform cases illustrating typical errors in GEDI L2A RH98 and corresponding CSDWC corrections. In (c)–(e), the blue circles indicate the nominal 25-m diameter GEDI footprints, and each high-resolution background image covers an 80×80 m ground area.

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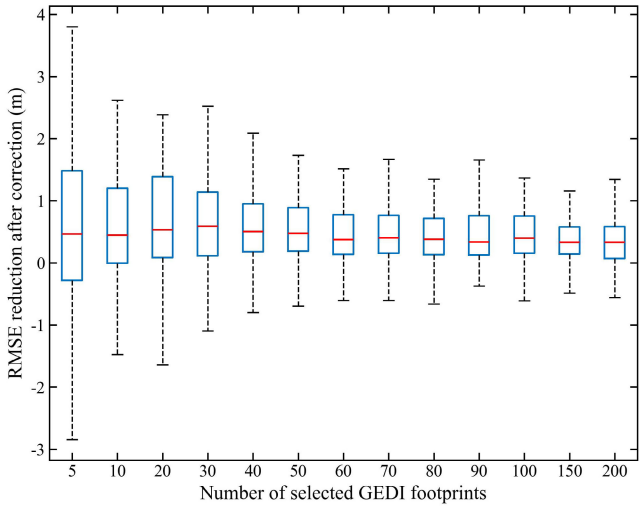


Fig. 10. Sensitivity of track-level elevation correction to the number of selected GEDI footprints. Blue boxes indicate the interquartile range, red lines denote the median, and black dashed whiskers extend to the most extreme nonoutlier values within 1.5 times the interquartile range.

B. Mechanistic Advantages and Performance Improvements of CSDWC

Previous studies have primarily attributed inaccuracies in GEDI L2A RH metrics to factors such as geolocation error [31] and topographic effects [38]. However, even after standard quality filtering and geolocation correction, we observed persistent and pronounced noise in canopy height changes derived from RH metrics-based differencing. If used directly as model inputs, such residual noise would substantially degrade subsequent mapping performance. Detailed inspection of the waveform profiles corresponding to these outliers [see Fig. 9(c)–(e)] indicates that the intrinsic limitations of the L2A height retrieval procedure constitute a dominant error source, an issue that has received limited attention in previous studies.

The noisy outliers produced by RH metrics-based differencing can be primarily explained by two mechanisms. First, in dense forests, strong coupling between canopy and ground signals complicates reliable ground elevation detection, which directly affects canopy height estimation [45]. Second, weak canopy returns coupled with background noise often prevent the L2A algorithm from correctly identifying the true canopy top, resulting in partial detection or complete omission of canopy signals.

Errors in ground elevation estimation occur predominantly in dense forests, where RH98 is highly sensitive to ground localization. Because RH98 is defined as the vertical distance between the detected ground and the upper canopy returns, any misidentification of the ground is directly propagated into canopy height estimates [35], [46], [47]. As illustrated in Fig. 9(c), ALS-derived simulated waveforms clearly indicate the true ground position, whereas GEDI ground returns are often mixed with surrounding signals. Under such conditions, the L2A algorithm may incorrectly select a stronger peak at a higher elevation as the ground return, thereby leading to systematic underestimation of canopy height change.

Errors in canopy signal extraction can be further categorized into incomplete detection and nondetection. In cases of incomplete detection [see Fig. 9(d)], only a portion of the canopy signal is identified, while residual returns above the detected RH98 remain unresolved due to noise coupling. When canopy signals are extremely weak or strongly contaminated by noise [see Fig. 9(e)], the L2A algorithm may fail to detect any canopy return and instead misclassify the ground peak as the dominant signal. Such situations commonly occur in sparse or fragmented forests and near forest edges [48], where GEDI footprints sample partial canopy elements without a continuous vertical structure. Under these conditions, weak canopy returns are easily confounded with background noise, resulting in high uncertainty in conventional RH98 extraction [49].

In contrast, the CSDWC algorithm is specifically designed to address these limitations. Statistical filtering of top-of-canopy noise delineates potential canopy regions, while ALS-derived simulated waveforms provide prior information on the expected canopy signal structure. This combined strategy improves the identification of weak canopy returns mixed with noise, thereby substantially enhancing the reliability of canopy height change estimation under difficult observation conditions.

C. Methodological Adaptability Across Forest Typologies

Waveform structure is closely related to species morphology and stand spatial structure [45], [50]. In sparse coniferous environments, the conical crown architecture and discontinuous canopy cover may cause space-borne footprints to intersect only the edges of adjacent crowns. Under this geometric configuration, the laser pulse may intercept upper foliage without encountering substantial woody biomass or trunk structures directly beneath the interception zone [e.g., Fig. 9(e)]. As a result, the middle and lower parts of the vertical profile may contain few valid returns, producing high radiometric noise and severe signal ambiguity. Under these conditions, the CSDWC algorithm shows strong correction capability. By using historical simulated waveforms as structural priors, the method anchors the retrieval process to the highest valid canopy return, thereby bypassing severe structural noise and mitigating the signal omission commonly associated with standard threshold-based algorithms.

In contrast, dense broadleaf deciduous forests present a different set of observational challenges, characterized by continuous and strongly coupled upper-canopy layers. Although dense foliage produces a strong initial canopy return, it also causes substantial energy attenuation that can obscure the underlying ground peak. In addition, the irregular and tightly packed broadleaf crowns often lead to top-signal coupling, making precise delineation of the absolute canopy top more difficult. In these dense ecosystems, the main mechanistic advantage of the CSDWC algorithm lies in its direct waveform differencing scheme. By focusing on the relative vertical displacement of waveform leading edges, the method largely avoids dependence on accurate ground elevation extraction. This waveform-level differencing therefore reduces the systematic underestimation caused by ground misidentification

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and enables robust height change estimation even when the lower part of the vertical profile is poorly resolved due to energy attenuation.

D. Implications for Large-Scale Forest Change Mapping

The current paradigm for large-scale canopy height change mapping mainly relies on map-level differencing, in which two independently modeled wall-to-wall height maps are subtracted [27], [28]. This approach is often constrained by considerable uncertainty introduced during spatial upscaling. According to the principle of error propagation, the uncertainty of the derived change $\sigma_{\Delta h}$, is determined by the root sum square of the individual bi-temporal model errors, expressed as $\sigma_{\Delta h} = (\sigma_{T1}^2 + \sigma_{T2}^2)^{1/2}$, where σ_{T1} and σ_{T2} represent the independent modeling errors associated with the first and second observation epochs. Consequently, independently upscaling canopy heights from two epochs can lead to cumulative observation and modeling errors. Such error accumulation may limit the ability to detect fine-scale forest dynamics and may obscure subtle structural degradation or growth.

The high-fidelity footprint-level height changes derived from the CSDWC method offer a robust alternative for regional and global forest monitoring. By providing highly accurate, directly measured vertical dynamics at the footprint scale, these estimations serve as superior training samples for machine learning models. When synergized with continuous spatial predictors such as optical time-series data, the canopy height change can be modeled directly in a single upscaling step. This direct modeling paradigm effectively mitigates the mathematical error accumulation associated with bi-temporal map subtraction. By minimizing the upscaling error to a single modeling phase, the proposed framework substantially reduces regional uncertainties, thereby providing a highly reliable data foundation for global carbon sink accounting, ecological restoration assessment, and climate change mitigation strategies.

E. Limitations and Future Directions

Several limitations remain in the proposed framework. First, a key assumption underlying the two-stage geolocation optimization algorithm is the relative stability of terrain morphology between the acquisition epochs. While generally valid in forested regions, violations of this assumption may introduce additional uncertainty. Future work could incorporate forest structural features as auxiliary alignment constraints. Although canopy height and physiological attributes may vary over time, tree spatial distributions are typically more stable and could provide robust anchors for waveform alignment.

Second, although the algorithm effectively captures bidirectional canopy height dynamics and demonstrates high consistency with reference data during both canopy growth and reduction phases [as evidenced in Fig. 9(b)], it relies on airborne simulated waveforms as prior information. This dependence restricts its operational autonomy; thus, future research should focus on developing more autonomous, prior-independent waveform-based height extraction algorithms, potentially leveraging generative techniques like diffusion models [51] to characterize complex waveforms coupled with background noise.

Third, the current framework has been validated in four study areas. Future studies could expand its application and evaluate its performance at continental or global scales.

VI. Conclusion

This study presents a footprint-level framework for forest canopy height change extraction by integrating GEDI observations with airborne LiDAR-simulated waveforms. By combining a two-stage geolocation correction strategy with the CSDWC-based waveform differencing method, the proposed framework significantly improves the accuracy of canopy height change retrieval. The main conclusions are as follows.

1) The integration of space-borne GEDI data with airborne LiDAR provides an effective and cost-efficient approach for fine-scale monitoring of forest canopy height changes.
2) The proposed two-stage optimization strategy, which combines DTM matching and ground-return matching, effectively corrects GEDI systematic geolocation errors and enhances the accuracy of footprint-level canopy height change estimation.
3) Converting airborne LiDAR point clouds into GEDI-like waveforms reduces systematic discrepancies between spaceborne waveform observations and airborne discrete-return measurements, thereby mitigating intersensor bias.
4) The proposed CSDWC method improves the detection of weak canopy signals embedded in background noise and demonstrates substantially higher reliability than the conventional GEDI L2A RH metrics-based differencing approach.

Future work will focus on extending this framework to broader ecological applications, such as aboveground biomass change monitoring. In addition, improving the robustness of canopy height change retrieval requires the development of standalone GEDI-based methods that do not rely on auxiliary airborne LiDAR data. Potential directions include the use of data augmentation techniques, such as diffusion-based methods, and the integration of deep learning architectures to better characterize complex waveform signals and spatiotemporal canopy dynamics.

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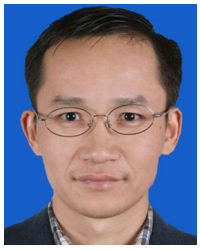
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